PART-A

****

1. Write a program to create a 3 X 3 matrices A and B and perform the following operations

a. $A^T.B$
b. $B^T.(A.A^T)$
c. $(A.A^T).B^T$
d. $[(B.B^T)+(A.A^T)-100I_3]^{-1}$

Date: 06-03-2026

****

```
A=matrix(1:9, nrow=3, ncol=3)
B=matrix(10:18, nrow=3, ncol=3)
I=diag(x=3)

qa<-t(A)%*%B
qb<-t(B)%*%(A%*%t(A))
qc<-A%*%t(A)%*%t(B)
qd<-solve((B%*%t(B))+(A%*%t(A))-100*I)

cat("Matrix A:")
print(A)
print(B)
print(qa)
print(qb)
print(qc)
print(qd)
```

Output:

```
print(A)
     [,1] [,2] [,3]
[1,]    1    4    7
[2,]    2    5    8
[3,]    3    6    9

> print(B)
     [,1] [,2] [,3]
[1,]   10   13   16
[2,]   11   14   17
[3,]   12   15   18

> print(qa)
     [,1] [,2] [,3]
[1,]   68   86  104
[2,]  167  212  257
[3,]  266  338  410
```

```
> print(qb)
     [,1] [,2] [,3]
[1,] 2598 3099 3600
[2,] 3300 3936 4572
[3,] 4002 4773 5544

> print(qc)
     [,1] [,2] [,3]
[1,] 3114 3348 3582
[2,] 3717 3996 4275
[3,] 4320 4644 4968

> print(qd)
             [,1]         [,2]         [,3]
[1,] -0.007214824  0.003191620  0.003598064
[2,]  0.003191620 -0.006516417  0.003775547
[3,]  0.003598064  0.003775547 -0.006046970
>
```

****

2. Write R program to find roots of quadratic equation using user defined function. Test the program user supplied values for all possible cases

Date: 06-03-2026

****

```
quadratic=function(a,b,c) {
   d=b^2-(4*a*c)
   if(d>0){
      root1=(-b+sqrt(d))/(2*a)
      root2=(-b-sqrt(d))/(2*a)
      return(c(round(root1,5), "and", round(root2,5)))
   }else if(d==0){
       root1=root2=((-b)/(2*a))
       return(c(round(root1,5),"and",round(root2,5)))
   }else{
      return(paste("No real Roots"))
  }
}

a=quadratic(-2,-3,-4)
cat("\nRoots are\t", a ,"\n")

b=quadratic(1,5,6)
cat("\nRoots are\t",b,"\n")

c=quadratic(1,6,9)
cat("\nRoots are\t",c,"\n")
```

Output:

```
> a=quadratic(-2,-3,-4)
> cat("\nRoots are\t", a ,"\n")

Roots are       No real Roots

> b=quadratic(1,5,6)
> cat("\nRoots are\t",b,"\n")

Roots are       -2 and -3
> c=quadratic(1,6,9)
> cat("\nRoots are\t",c,"\n")

Roots are       -3 and -3
```

****

3. Write R script to generate prime numbers between two numbers using loops

Date: 13-03-2026

****

```
m=1
n=20
prime=c()
for(i in seq(m,n)){      # 1,2,3,4,5…20
   if(i==2)  {
       prime=c(i)
       next
    }
    flag=1
    for(j in seq(2, i/2))  {     # 2,3,4….. i/2
      if((i%%j)==0)    {
          flag=0
          break
      }
    }
    if(flag==1)
       prime=c(prime,i)
}
print(prime)
```

Output:

```
[1]  2  3  5  7 11 13 17 19
```

****

4. Write an R program to create a list containing strings, numbers, vectors and logical values and do the following manipulations over the list

a. Access the first element in the list

b. Give the names to the elements in the list

c. Add element at some positions in the list

d. Remove the element

e. print the first and third element

f. Update the third element

Date: 13-03-2026

****

#create a list containing strings, numbers, vectors and logical values

```
l<-list(c("St1","St2","St3"), c(1,2,3,4), c(3,3.4), c(T,F,T,F))

#a.Access the first element in the list
l[1]

#b.Give the names to the elements in the list
names(l)<-c("St","Int","Num","Logic")

#c. Add element at some positions in the list
l$f<-c("red","Blue")                    #Name is f

#d. Remove the element
l[-1]

#e. print the first and third element
l[c(1,3)]

#f. Update the third element
l[3]<-7
l                                #Print List
```

Output:

```
> #create a list containing strings, numbers, vectors and logical values
> l<-list(c("St1","St2","St3"),c(1,2,3,4),c(3,3.4),c(T,F,T,F))

> #a.Access the first element in the list
> l[1]
[[1]]
[1] "St1" "St2" "St3"

> #b.Give the names to the elements in the list
> names(l)<-c("St","Int","Num","Logic")

> #c. Add element at some positions in the list
> l$f<-c("red","Blue")
```

```
> #d. Remove the element
> l[-1]
$Int
[1] 1 2 3 4

$Num
[1] 3.0 3.4

$Logic
[1]  TRUE FALSE  TRUE FALSE

$f
[1] "red"  "Blue"


> #e. print the first and third element
> l[c(1,3)]
$St
[1] "St1" "St2" "St3"

$Num
[1] 3.0 3.4


> #f. Update the third element
> l[3]<-7

> l
$St
[1] "St1" "St2" "St3"

$Int
[1] 1 2 3 4

$Num
[1] 7

$Logic
[1]  TRUE FALSE  TRUE FALSE

$f
[1] "red"  "Blue"
```

****

5. The following table shows the time taken (in minutes) by 100 students to travel to school on a particular day.

Time	0-5	5-10	10-15	15-20	20-25
No. of students	5	25	40	17	13

a. Draw the histogram

b. Draw frequency polygon

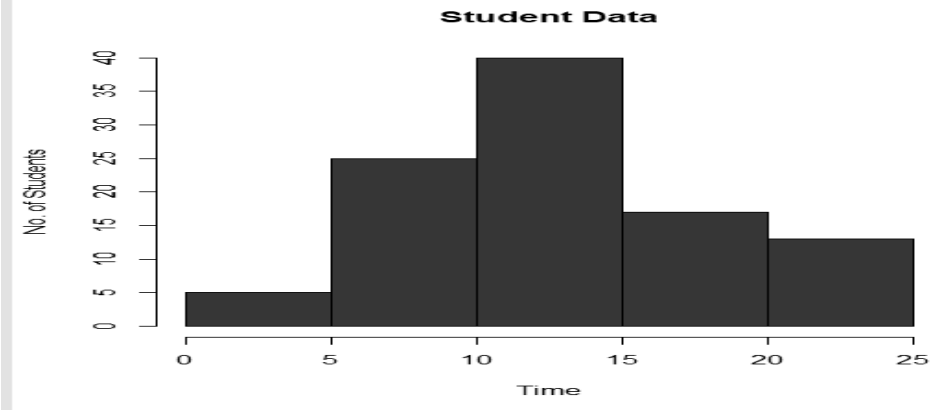
Date: 13-03-2026

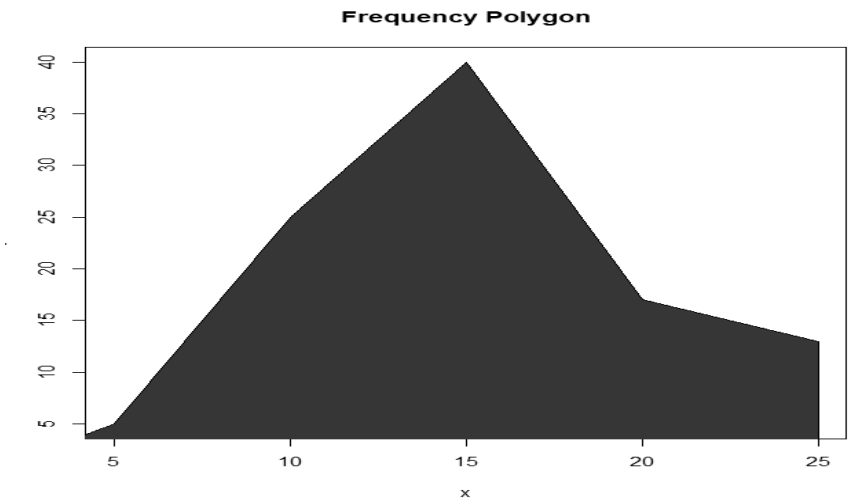
****

#Histogram

```
l= c(0,5,10,15,20)
u=c(5,10,15,20,25)
mid=(l+u)/2
f=c(5,25,40,17,13)
y=rep(mid,f)              #  repetition of mid f times  y=2.5(5 times),7.5 (25 times), 12.5(40 times)..etc

brk=seq(0,25,5)                # 0, 5, 10, 15, 20, 25      # breaks specify the custom breaks- bar width
hist(y, breaks=brk, main="Student Data",   xlab="Time",   ylab="No. of Students",   col="red")
axis(side=2, at=seq(5,40,5))          #In Y axis, the values 0, 5, 10, 15,……,40 marked, main = title
```

```
#Frequency polygon
x<-seq(5,25,5)                    #5, 10, 15, 20, 25
y<-c(5,25,40,17,13)               #5, 25, 40, 17, 13
plot(x, y, type="l", main="Frequency Polygon")          #type l means lines
polygon(c(1, x, 25), c(0,y,0),col="red")        #  1, 5, 10, 15, 20, 25, 25          0, 5, 25, 40, 17, 13, 0
```





****

6. Write an R program to create a Data Frame with following details and do the following operations.

ItemCode	itemCategory	ItemPrice
1001	Electronics	700
1002	Desktop Supplies	300
1003	Office Supplies	350
1004	USB	400
1005	CD Drive	800

a. Subset the Data frame and display the details of only those items whose price is greater than or equal to 350.

b. Subset the Data frame and display only the items where the category is either "Office Supplies" or "Desktop Supplies"

c. Subset the Data frame and display the items where the Itemprice between 300 and 700

d. Compute the sum of all ItemPrice

e. Create another Data Frame called "item-details" with three different fields itemCode, ItemQtyonHand and ItemReorderLvl and merge the two frames

Date: 08-04-2026

****

#create data frame

```
myd<-data.frame(ItemCode=c(1001,1002,1003,1004,1005),
ItemCategory=c("Electronics","Desktop Supplies","office Supplies","USB","CD Drive"),
ItemPrice=c(700,300,350,400,800))
```

#display data frame

```
myd
```

#a. Subset the Data frame and display the details of only those items whose price is greater than or equal to 350.

```
myd[myd$ItemPrice>=350,]
```

#b. Subset the Data frame and display only the items where the category is either "Office Supplies" or "Desktop Supplies"

```
myd[myd$ItemCategory=="Office Supplies" | myd$ItemCategory=="Desktop Supplies",]
```

#c. Subset the Data frame and display the items where the Itemprice between 300 and 700

```
myd[myd$ItemPrice>=300 & myd$ItemPrice<=700,]
```

#d.Compute the sum of all ItemPrice

```
sum(myd$ItemPrice)
```

#e. **Create another Data Frame called "item-details" with three different fields itemCode, ItemQtyonHand and ItemReorderLvl and merge the two frames**

```
Itemdetails<-data.frame(ItemCode=c(1001,1002,1003,1004,1005),
                        ItemQtyonHand=c(2,3,4,5,4),
                        ItemRecorderLvl=c(1,2,3,2,3))

mer<-cbind(myd, Itemdetails)
cat("Merged frames")
mer
```

OUTPUT:

#create data frame

```
> myd<-data.frame(ItemCode=c(1001,1002,1003,1004,1005),
+ ItemCategory=c("Electronics","Desktop Supplies","office Supplies","USB","CD Drive"),
+ ItemPrice=c(700,300,350,400,800))
```

> #display data frame

```
> myd

  ItemCode   ItemCategory       ItemPrice
1    1001     Electronics          700
2    1002     Desktop Supplies     300
3    1003     office Supplies      350
4    1004     USB                  400
5    1005     CD Drive             800
```

> #a. Subset the Data frame and display the details of only those items whose price is greater than or equal to 350.

```
> myd[myd$ItemPrice>=350,]

  ItemCode    ItemCategory   ItemPrice
1    1001     Electronics        700
3    1003     office Supplies    350
4    1004          USB           400
5    1005       CD Drive         800
```

> #b. Subset the Data frame and display only the items where the category is either "Office Supplies" or "Desktop Supplies"

```
> myd[myd$ItemCategory=="Office Supplies"|myd$ItemCategory=="Desktop Supplies",]
  ItemCode     ItemCategory ItemPrice
2    1002 Desktop Supplies       300
3    1003      office Supplies      350
```

> #c. Subset the Data frame and display the items where the Itemprice between 300 and 700

> myd[myd$ItemPrice>=300&myd$ItemPrice<=700,]

```
  ItemCode     ItemCategory    ItemPrice
```

```
1    1001    Electronics        700
2    1002    Desktop Supplies   300
3    1003    office Supplies    350
4    1004        USB            400
```

> #d.Compute the sum of all ItemPrice

```
> sum(myd$ItemPrice)
[1] 2550
```

> #e. Create another Data Frame called "item-details" with three different fields itemCode, ItemQtyonHand and ItemReorderLvl and merge the two frames

```
> Itemdetails<-data.frame(ItemCode=c(1001,1002,1003,1004,1005),
+                         ItemQtyonHand=c(2,3,4,5,4),
+                         ItemRecorderLvl=c(1,2,3,2,3))
> merge<-cbind(myd,Itemdetails)
> #Merged data frames

> merge
  ItemCode    ItemCategory  ItemPrice ItemCode ItemQtyonHand ItemRecorderLvl
1    1001    Electronics        700     1001          2              1
2    1002    Desktop Supplies   300     1002          3              2
3    1003    office Supplies    350     1003          4              3
4    1004    USB                400     1004          5              2
5    1005    CD Drive           800     1005          4              3
      >>
```

****

7. Create a factor marital_status with levels Married, single, divorced. Perform the following operations on this factor

a. Check the variable is a factor

b. Access the 2nd and 4th element in the factor

c. Remove third element from the factor

d. Modify the second element of the factor

e. Add new level widowed to the factor and add the same level to the factor marital_status

Date: 08-04-2026

****

#Creating factor

fac=factor(c("married","single","divorced","married","single"),levels=c("married","single","divorced"))

#Display factor

Fac

 # Check the variable is a factor

is.factor(fac)

#Access the 2nd and 4th element in the factor

fac[c(2,4)]

#Remove third element from the factor

fac[-3]

#Modify the second element of the factor

fac[2]="married"

fac

#Add new level widowed to the factor and add the same level to the factor marital_status

levels(fac)=c(levels(fac),"widow")

levels(fac)

<u>OUTPUT:</u>

```
#Creating factor
>
fac=factor(c("married","single","divorced","married","single"),levels=c("married","single","divorced"))
> #Display factor
> fac
[1] married  single   divorced married  single
Levels: married single divorced
> # Check the variable is a factor
> is.factor(fac)
[1] TRUE
> #Access the 2nd and 4th element in the factor
> fac[c(2,4)]
```

```
[1] single  married
Levels: married single divorced
> #Remove third element from the factor
> fac[-3]
[1] married single  married single
Levels: married single divorced
> #Modify the second element of the factor
> fac[2]="married"
> fac
[1] married  married  divorced married  single
Levels: married single divorced
> #Add new level widowed to the factor and add the same level to the factor marital_status
> levels(fac)=c(levels(fac),"widow")
> levels(fac)
[1] "married"  "single"   "divorced" "widow"
>
```

****

8. Write a R language Script for following operation on Iris Data Set

1. Load the Iris Dataset

2. View first six rows of iris dataset

3. Summarize iris dataset

4. Display number of rows and columns

5. Display column names of dataset.

6. Create histogram of values for sepal length

7. Create scatterplot of sepal width vs. sepal length

8. Create boxplot of sepal width vs. sepal length

9. Find Pearson correlation between Sepal.Length and Petal.Length

10. Create correlation matrix for dataset

Date: 08-04-2024

****

```
data("iris")
head(iris)
summary(iris)
dim(iris)
names(iris)
hist(iris$Sepal.Length, col="blue", xlab="Sepal Length",ylab="frequency", main="Histogram")
plot(iris$Sepal.Width, iris$Sepal.Length, col="red", cex=2, xlab="sepal width", ylab="sepal length", main="scatterplot")
boxplot(Sepal.Length~Species,data=iris,main="Box Plot",xlab="Species",ylab="Scpal length")
cor(iris$Sepal.Length,iris$Petal.Length,method="pearson")
cor(iris[1:4])
```

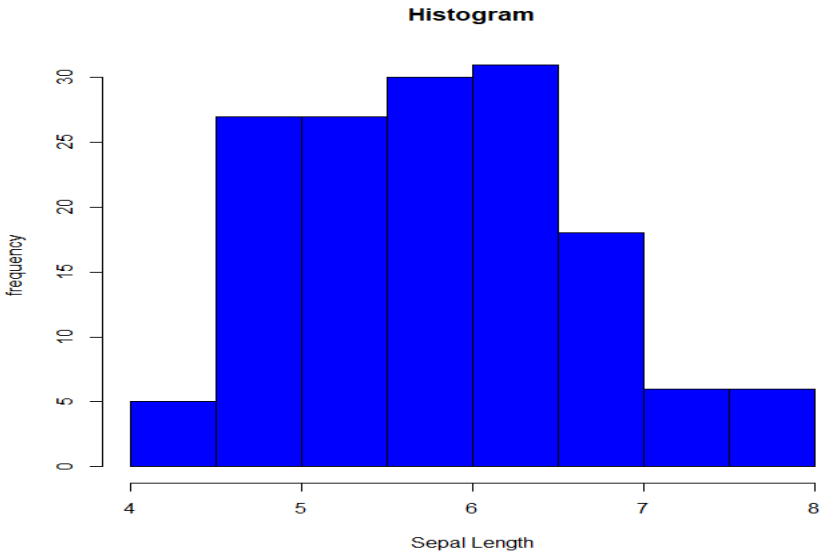
OUTPUT:

```
> cat("1.Loading IRIS Data Set")
1.Loading IRIS Data Set
> data("iris")

> cat("2.first six rows of iris dataset ")
2.first six rows of iris dataset
 > head(iris)
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa

> cat("3.Summary")
3.Summary
> summary(iris)
  Sepal.Length    Sepal.Width     Petal.Length    Petal.Width
 Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100
 1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300
 Median :5.800   Median :3.000   Median :4.350   Median :1.300
```

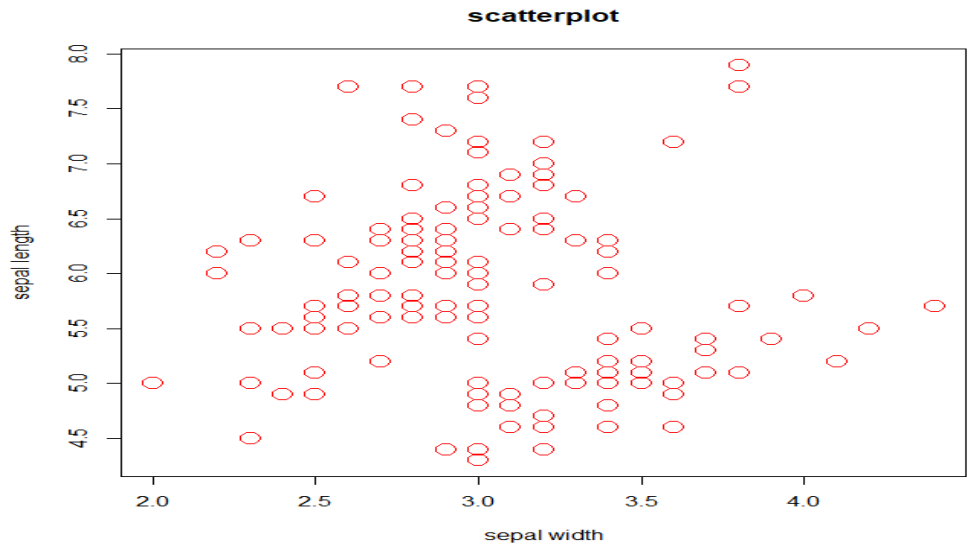
```
 Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
 3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
 Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500
       Species
 setosa    :50
 versicolor:50
 virginica :50

> cat("4.Dimension")
4.Dimension
> dim(iris)
[1] 150   5

> #5.column names of dataset
> names(iris)
[1] "Sepal.Length" "Sepal.Width"  "Petal.Length" "Petal.Width"  "Species"
>
```

#6.Hisogram of values for sepal length

```
> hist(iris$Sepal.Length, col="blue",  xlab="Sepal Length",ylab="frequency", main="Histogram")
```



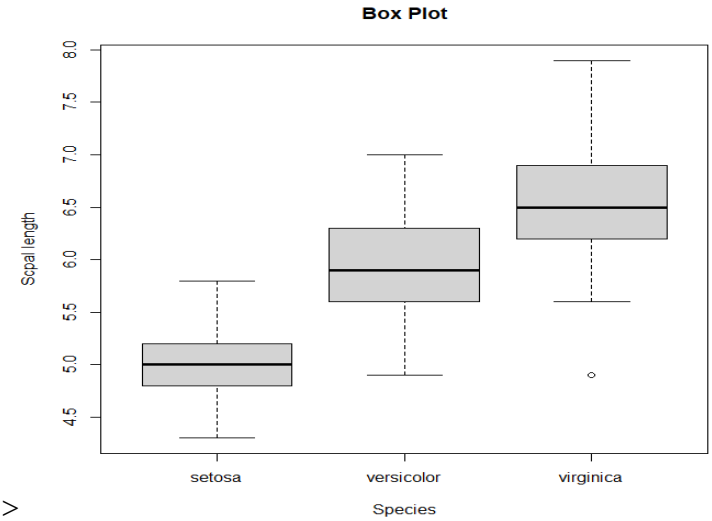

```
> #7.scatterplot of sepal width vs. sepal length   ; cex=char expansion
> plot(iris$Sepal.Width, iris$Sepal.Length, col="red", cex=2, xlab="sepal width", ylab="sepal
length",main="scatterplot")
```



>

> #8.boxplot of sepal width vs. sepal length x~y x grouped by the levels of the y variable.

> boxplot(Sepal.Length~Species, data=iris, main="Box Plot", xlab="Species",ylab="Scpal length")



>

> #9. Find Pearson correlation between Sepal.Length and Petal.Length

> cor(iris$Sepal.Length,iris$Petal.Length,method="pearson")

[1] 0.8717538

> #10. correlation matrix for dataset

> cor(iris[1:4])

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
Sepal.Length	1.0000000	-0.1175698	0.8717538	0.8179411
Sepal.Width	-0.1175698	1.0000000	-0.4284401	-0.3661259
Petal.Length	0.8717538	-0.4284401	1.0000000	0.9628654
Petal.Width	0.8179411	-0.3661259	0.9628654	1.0000000

- - - -

PART-B

****

1. Write a R program to create a Vector containing following 8 values and perform the following operations.

4 3 0 5 2 9 4 5

a. Find mean, median, mode.

b. Find the range.

c. Find the 35th and 78th percentile.

d. Find the variance and standard deviation

e. Find the interquartile range.

f. Find the z-score for each value.

Date: 17-04-2026

****

```
# Create the vector with the given values
vec <- c(4, 3, 0, 5, 2, 9, 4, 5)

# a. Find mean, median, mode
mean_vec <- mean(vec)
median_vec <- median(vec)
counts <- table(vec)            # 0 2 3 4 5 9   #table function generates frequency count, names-counts
                                # 1 1 1 2 2 1

# Display repeated numbers and their counts
repeated_numbers <- names(counts[counts > 1])   # 4, 5 Take names in counts array counts >1 entries
for (num in repeated_numbers) {
        count <- counts[num]
        cat("Mode:", num, "\n")
}

# b. Find the range
range_vec <- max(vec) - min(vec)          # 9

# c. Find the 35th and 78th percentile
per_vec <- quantile(vec,c(0.35,0.78))          # 3.45    5.00

# d. Find the variance and standard deviation
variance_vec <- var(vec)
std_dev_vec <- sd(vec)

# e. Find the interquartile range
iqr_vec <- IQR(vec)

# f. Find the z-score for each value
z_scores <- (vec - mean_vec) / std_dev_vec

# Print the results
cat("a. Mean:", mean_vec, "\n")
cat("   Median:", median_vec, "\n")

cat("b. Range:", range_vec, "\n\n")
cat("c. 35th and 78thPercentile:", per_vec, "\n\n")
```

```
cat("d. Variance:", variance_vec, "\n")
cat("   Standard Deviation:", std_dev_vec, "\n\n")

cat("e. Interquartile Range:", iqr_vec, "\n\n")

cat("f. Z-Scores:\n")
print(z_scores)
```

OUTPUT:

```
> # Create the vector with the given values
> vec <- c(4, 3, 0, 5, 2, 9, 4, 5)
>
> # a. Find mean, median, mode
> mean_vec <- mean(vec)
> median_vec <- median(vec)
>
> counts <- table(vec)
>
> # Display repeated numbers and their counts
> repeated_numbers <- names(counts[counts > 1])
> for (num in repeated_numbers) {
+   count <- counts[num]
+   cat("Mode:", num, "\n")
+ }
Mode: 4
Mode: 5
>
> # b. Find the range
> range_vec <- max(vec) - min(vec)
>
> # c. Find the 35th and 78th percentile
> per_vec <- quantile(vec,c(0.35,0.78))
>
> # d. Find the variance and standard deviation
> variance_vec <- var(vec)
> std_dev_vec <- sd(vec)
>
> # e. Find the interquartile range
> iqr_vec <- IQR(vec)
>
> # f. Find the z-score for each value
> z_scores <- (vec - mean_vec) / std_dev_vec
>
> # Print the results
> cat("a. Mean:", mean_vec, "\n")
a. Mean: 4
> cat("   Median:", median_vec, "\n")
   Median: 4
>
> cat("b. Range:", range_vec, "\n\n")
b. Range: 9
>
> cat("c. 35th and 78thPercentile:", per_vec, "\n\n")
c. 35th and 78thPercentile: 3.45 5
```

```
> cat("d. Variance:", variance_vec, "\n")
d. Variance: 6.857143
> cat("   Standard Deviation:", std_dev_vec, "\n\n")
   Standard Deviation: 2.618615

>
> cat("e. Interquartile Range:", iqr_vec, "\n\n")
e. Interquartile Range: 2.25

>
> cat("f. Z-Scores:\n")
f. Z-Scores:
> print(z_scores)
[1]  0.0000000 -0.3818813 -1.5275252  0.3818813 -0.7637626  1.9094065  0.0000000
[8]0.3818813
```

Solution:

X	X(Ascending)	X_i-X'	$(X_i-X')^2$
4	0	-4	16
3	2	-2	4
0	3	-1	1
5	4	0	0
2	4	0	0
9	5	1	1
4	5	1	1
5	9	5	25
	∑X=32		$\sum(Xi-X')^{2=}48$

a. **Mean** X' = ∑X/n = 32/8 = 4 (n=8)

Median = Average of middle two numbers = (4+4)/2 = 4

Mode = The value that appears more often = 4 and 5

b. **Range** = Difference between largest and smallest value = 9-0 = 9

c. **35th and 78th percentile**

I = P/100(N), I = 35/100(8), I = 2.8 is not whole number.
So whole no. part of I+1 = 2.8 + 1 = 3.8 = 4
=4th element = 4.

I = P/100(N), I = 78/100(8), I = 6.24 is not whole number.
So whole no. part of i+1=6.24+1=7.24=7
= 7th element = 5.

d. **Variance and standard deviation**

$S^2=\sum(X_i-X')^2/(n-1)$
=48/8-1=6.85741
$S=\sqrt{\sum(X_i-X')^2/(n-1)}$
$=\sqrt{6.85741}$
=2.618614

e. **Interquartile range.**

Q_3-Q_1

Q3 = 75/100(n) = 75/100(8) = 6 Average of Ith and $I+1^{th}$ element = (5+5)/2 = 5

Q1 = 25/100(n) = 25/100(8) = 2 Average of Ith and $I+1^{th}$ element = (2+3)/2 = 2.5

f. **z-score for each value.**

Z = (Xi-X')/S

= (0-4)/2.618614 = -1.5275

= (2-4)/2.618614 = -0.7637

= (3-4)/2.618614 = -1.3818

= (4-4)/2.618614 = 0

= (4-4)/2.618614 = 0

= (5-4)/2.618614 = 0.38188

= (5-4)/2.618614 = 0.38188

= (9-4)/2.618614 = 1.90940

****

2. Write R script to find the correlation coefficient and type of correlation between advertisement expenses and sales volume using Karl Pearson's coefficient of correlation method (Direct Method).

Firm	1	2	3	4	5	6	7	8	9	10
Advertisement Exp. (Rs. In Lakhs)	11	13	14	16	16	15	15	14	13	13
Sales Volume (Rs. In Lakhs)	50	50	55	60	65	65	65	60	60	50

Date: 17-04-2024

****

R-Code:

```
# Example data (replace with your actual data)
ad_ex<- c(11,13,14,16,16,15,15,14,13,13)
   s_v <- c(50,50,55,60,65,65,65,60,60,50)
# Calculate the correlation coefficient
cor_cf <- cor(ad_ex, s_v, method = "pearson")
# Determine the type of correlation
if (cor_cf> 0)
  {
 cor_type <- "Positive correlation"
}else if (cor_cf < 0)
  {
 cor_type <- "Negative correlation"
 }else
 {
 cor_type <- "No correlation"
}
# Print the results
cat("Correlation Coefficient:", cor_cf, "\n")
cat("Type of Correlation:", cor_type, "\n")
```

<u>Output:</u>

```
# Example data (replace with your actual data)
> ad_ex<- c(11,13,14,16,16,15,15,14,13,13)
> s_v <- c(50,50,55,60,65,65,65,60,60,50)
>
> # Calculate the correlation coefficient
> cor_cf <- cor(ad_ex,s_v, method = "pearson")
>
> # Determine the type of correlation
> if (cor_cf> 0)
+   {
+  cor_type <- "Positive correlation"
```

```
+ }else if (cor_cf < 0)
+    {
+   cor_type <- "Negative correlation"
+  }else
+  {
+   cor_type <- "No correlation"
+ }
>
> # Print the results
> cat("Correlation Coefficient:", cor_cf, "\n")
Correlation Coefficient: 0.7865665
> cat("Type of Correlation:", cor_type, "\n")
Type of Correlation: Positive correlation
>
```

Manual solution:

Karl Pearson's coefficient of correlation $r = \frac{N\sum XY - \sum X.\sum Y}{\sqrt{N\sum X^2 - (\sum X)^2 . N\sum Y^2 - (\sum Y)^2}}$

X	Y	X^2	Y^2	X*Y
11	50	121	2500	550
13	50	169	2500	650
14	55	186	3025	770
16	60	256	3600	960
16	65	256	4225	1040
15	65	225	4225	975
15	65	225	4225	975
14	60	196	3600	840
13	60	169	3600	780
13	50	169	2500	650
$\Sigma x = 140$	$\Sigma y = 580$	Σx^2=1982	$\Sigma y^2 = 34000$	$\Sigma xy = 8190$

Substitute to the formula we get,

$$r = \frac{10 * 8190 - (140 * 580)}{\sqrt{[(10 * 1982) - (140)^2] * [(10 * 34000) - (580)^2}}$$

$$= \frac{81900 - 81200}{\sqrt{220 * 3600}}$$

$$= \frac{700}{\sqrt{792000}}$$

$$= \frac{700}{889.943818}$$

Therefore correlation coefficient using Karl Pearson's coefficient correlation method

r =0.786566

0.786566>0

So there is a positive correlation between advertise expenses and sales volume.

****

3. Write R script to compute the regression equation of y on x from the following data. Predict the value of y when x=7

X	2	4	5	6	8	11
Y	18	12	10	8	7	5

Date: 17-04-2024

****

R Code:

```
3# Example data
x <- c(2,4,5,6,8,11)
y <- c(18,12,10,8,7,5)

# Fit a linear regression model
model <- lm(y ~ x)

# Get the regression equation
intercept <- coef(model)[1]                # y =mx+c    y = -1.34 x + 18.04
slope <- coef(model)[2]

# Predict the value of y when x = 7
x_new <- 7
y_predicted <- predict(model, data.frame(x = x_new))

# Print the regression equation
cat("Regression Equation: y =", round(slope, 2), "x +", round(intercept, 2),"\n")

# Print the predicted value of y when x = 7
cat("Predicted y when x =", x_new, "is approximately", round(y_predicted, 2), "\n")
```

OUTPUT:

```
> 3# Example data
[1] 3
> x <- c(2,4,5,6,8,11)
> y <- c(18,12,10,8,7,5)
>
> # Fit a linear regression model
> model <- lm(y ~ x)
>
> # Get the regression equation
> intercept <- coef(model)[1]
> slope <- coef(model)[2]
>
> # Predict the value of y when x = 7
> x_new <- 7
> y_predicted <- predict(model, data.frame(x = x_new))
```

```
> # Print the regression equation
> cat("Regression Equation: y =", round(slope, 2), "x +",round(intercept, 2),"\n")
Regression Equation: y = -1.34 x + 18.04
>  # Print the predicted value of y when x = 7
> cat("Predicted y when x =", x_new, "is approximately", round(y_predicted, 2), "\n")
Predicted y when x = 7 is approximately 8.66
>
```

Solution:

x	**y**	**x^2**	**∑xy**
2	18	4	36
4	12	8	48
5	10	25	50
6	8	36	48
8	7	64	56
11	5	121	55
∑x=36	$\sum y = 60$	$\sum x^2 = 266$	$\sum xy = 293$

$\bar{x} = \frac{36}{6} = 6$ [$\bar{x} = \frac{\sum x}{n}$]

$\bar{y} = \frac{60}{6} = 10$

<u>Regression equation of y on x</u>

$$y - \bar{y} = byx\ (x - \bar{x})$$

Where $byx = \frac{N\sum xy - \sum x.\sum y}{N\sum x^2 - (\sum x)^2}$

byx= $\frac{6(293) - 36(60)}{6(266) - 36^2}$

$= \frac{1758 - 2160}{1596 - 1296}$

$= \frac{-402}{300}$

$y - \bar{y} = by\ x\ (x - \bar{x})$

$y - 10 = -1.34(x - 6)$

$y = -1.34x + 8.04 + 10$

$y = -1.34x + 18.04$

Therefore, Regression equation of y on x

$\underline{y = -1.34x + 18.04}$

Value of y where x=7 is

Sub 7 to x

$y = -1.34(7) + 18.04$

$\underline{y = 8.66}$

****

4. The times taken by a large group of students to complete a piece of homework, T minutes, are Normally distributed with a mean of 57 minutes and standard deviation of 6.5. Find the probability that the time taken by a random student from the group to complete this homework will be less than 60 minutes.

Write R script to Find the probability that the time taken by a random student from the group to complete this homework

a) Will be less than 60 minutes

b) Between 50 and 80 minutes

Date: 24-04-2024

****

```
# Mean and standard deviation of the distribution
mean_time <- 57
std_dev <- 6.5

# a) Probability that the time taken is less than 60 minutes
prob_less_than_60 <- pnorm(60, mean_time, std_dev)
cat("Probability that time taken is less than 60 minutes:", round(prob_less_than_60, 4), "\n")

# b) Probability that the time taken is between 50 and 80 minutes
prob_between_50_and_80 <- pnorm(80, mean_time, std_dev) - pnorm(50, mean_time, std_dev)
cat("Probability that time taken is between 50 and 80 minutes:", round(prob_between_50_and_80, 4), "\n")
```

<u>OUTPUT:</u>

```
# Mean and standard deviation of the distribution
> mean_time <- 57
> std_dev <- 6.5
>
> # a) Probability that the time taken is less than 60 minutes
> prob_less_than_60 <- pnorm(60, mean_time, std_dev)
> cat("Probability that time taken is less than 60 minutes:", round(prob_less_than_60, 4), "\n")
Probability that time taken is less than 60 minutes: 0.6778
>
> # b) Probability that the time taken is between 50 and 80 minutes
> prob_between_50_and_80 <- pnorm(80, mean_time, std_dev) - pnorm(50, mean_time, std_dev)
> cat("Probability that time taken is between 50 and 80 minutes:", round(prob_between_50_and_80, 4), "\n")
Probability that time taken is between 50 and 80 minutes: 0.859
```

<u>Solution:</u>

Probability that the time taken is less than 60 minutes:

Given:

x=60

Mean=57

Standard deviation=6.5

z-score $z = \frac{x-\mu}{\sigma}$

$z = \frac{60-57}{6.5} = \frac{3}{6.5} = 0.46$

P(x<60)
P(z<1)
=(0.5 + P(z= 0.46))
=0.5+0.1772
Therefore the probability that the time taken is less than 60 minutes:=0.6772

b) Probability that the time taken is between 50 and 80 minutes:

Z_50 = $\frac{50-57}{6.5}$ = -1.07

Z_80 = $\frac{80-57}{6.5}$ = 3.53

=prob(x=80) – prob(x=50)
= p(z=3.5) - p(z=-1.0769)
= (0.5+0.4998) – (0.5 – 0.3577)
= 0.9998 – 0.1423
Therefore Probability that the time taken is between 50 and 80 minutes: = 0.8575

The Z-table gives the probability for **values less than a given Z-score** in a standard normal distribution.

- A Z-score of **0.46** corresponds to an area of **0.1772** in the right tail beyond the mean.
- The standard normal distribution is symmetric, with **50% (or 0.5) of the area to the left of the mean (Z = 0).**
- Since our Z-score is **positive (0.46),** it means we are looking for the total probability to the left of Z = 0.46.

Thus, we calculate:

$$P(Z < 0.46) = P(Z < 0) + P(0 < Z < 0.46)$$

From the standard normal table:

$$P(Z < 0) = 0.5$$

$$P(0 < Z < 0.46) = 0.1772$$

So, adding both values:

$$P(Z < 0.46) = 0.5 + 0.1772 = 0.6772$$

****

5. Write R script to perform the following using binomial distribution

i. If n=4 and p=0.10, find P(x=3)

ii. If n=12 and p=0.45, find P(5<=x<=7)

Date: 24-04-2024

****

R Code:

```
# Define the parameters
n1 <- 4
p1 <- 0.10
n2 <- 12
p2 <- 0.45

# Calculate P(x=3) for the first scenario
prob_x3 <- dbinom(3, size = n1, prob = p1)

# Calculate P(5<=x<=7) for the second scenario
prob_range <- sum(dbinom(5:7, size = n2, prob = p2))

# Print the results
cat("P(x=3) when n=4 and p=0.10:", prob_x3, "\n")
cat("P(5<=x<=7) when n=12 and p=0.45:", prob_range, "\n")
```

OUTPUT:

```
> # Define the parameters
> n1 <- 4
> p1 <- 0.10
> n2 <- 12
> p2 <- 0.45
>
> # Calculate P(x=3) for the first scenario
> prob_x3 <- dbinom(3, size = n1, prob = p1)
>
> # Calculate P(5<=x<=7) for the second scenario
> prob_range <- sum(dbinom(5:7, size = n2, prob = p2))
>
> # Print the results
> cat("P(x=3) when n=4 and p=0.10:", prob_x3, "\n")
P(x=3) when n=4 and p=0.10: 0.0036
> cat("P(5<=x<=7) when n=12 and p=0.45:", prob_range, "\n")
P(5<=x<=7) when n=12 and p=0.45: 0.583828
```

Solution:

$p(x) = n_{c_r} p^r q^{n-r}$ p=1-q

$n_{c_r} = \frac{n!}{r!(n-r)!}$ q=1-p

i. If n=4 and p=0.10, find P(x=3)

Given,

i) n=4 r=3 p=0.10

q = 1-p , q= 1-0.10, = <u>0.90</u>

Therfore,

p(x=3)= $4_{c_3}(0.10)^3(0.90)^{4-3}$

=> $p(x) = \frac{4!}{3!(4-3)!}(0.10)^3(0.90)^1$

=> $\frac{4*\cancel{3*2*1}}{\cancel{3*2*1}}(0.10)^3(0.90)$

=>$40.001 * 0.90$

=>**<u>0.0036</u>**

ii. If n=12 and p=0.45, find P(5<=x<=7)

$p(x = 5) + p(x = 6) + p(x = 7)$

$p(5 \leq x \leq 7)$

- $p(x = 5)$

$Given,\ \ n = 12\ \ p = 0.45\ \ r = 5$

q=1-0.45=0.55

Therefore, p(x=5)=$^{12}C_5(0.45)^5(0.55)^7$ []

$\frac{12!}{5!(12-5)!}(0.45)^5(0.55)^7$

$= \frac{12x11x10x9x8x7x6x5x4x3x2x1}{(5x4x3x2x1)x(7x6x5x4x3x2x1)}\ x(0.45)^5x(0.55)^7$

=792x(0.01845)x(0.015224)

<u>=0.222</u>

- P(x=6)

=$^{12}C_6(0.45)^6(0.55)^6$ (q=1-0.45=0.55)

$= \frac{12!}{6!(12-6)!}(0.45)^6(0.55)^6$

$= \frac{12x11x10x9x8x7x6x5x4x3x2x1}{(6x5x4x3x2x1)x(6x5x4x3x2x1)}\ x(0.45)^6\ x(0.55)^6$

=924x0.0083x0.0276

<u>=0.212</u>

- P(x=7)

=$^{12}C_7(0.45)^7(0.55)^5$ (q=1-0.45=0.55)

$= \frac{12!}{7!\ (12-7)!}(0.45)^7(0.55)^5$

$= \frac{12x11x10x9x8x7x6x5x4x3x2x1}{(7x6x5x4x3x2x1)\ x\ (5x4x3x2x1)}\ x\ (0.45)^7x\ (0.55)^5$

= 792 x 0.0037 x 0.05032

<u>= 0.147</u>

Therefore, P(5 <= x <= 7)

= 0.222 + 0.212 + 0.147

<u>= 0.581</u>

****

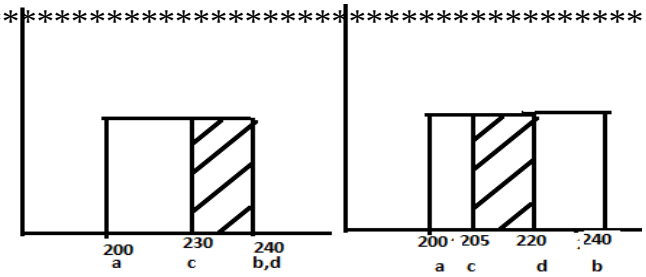
6. Perform the following using uniform distribution between 200 and 240

i. P(x>230)

ii. P(205≤x≤220)

Date: 24-04-2024

****



R Code:

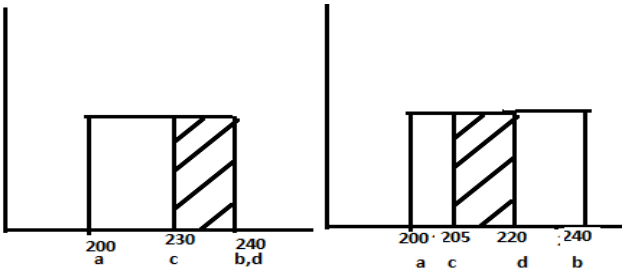
```
# Define the parameters for the uniform distribution
a <- 200  # Lower limit
b <- 240  # Upper limit

# Calculate P(x > 230)
prob_x_gt_230 <-  punif(240, min = a, max = b)- punif(230, min = a, max = b)
# Calculate P(205 ≤ x ≤ 220)
prob_range_205_to_220 <- punif(220, min = a, max = b) - punif(205, min = a, max = b)

# Print the results
cat("P(x > 230):", prob_x_gt_230, "\n")
cat("P(205 ≤ x ≤ 220):", prob_range_205_to_220, "\n")
```

OUTPUT:

```
> # Define the parameters for the uniform distribution
> a <- 200  # Lower limit
> b <- 240  # Upper limit
>
> # Calculate P(x > 230)
> prob_x_gt_230 <-  punif(240, min = a, max = b)- punif(230, min = a, max = b)
> # Calculate P(205 ≤ x ≤ 220)
> prob_range_205_to_220 <- punif(220, min = a, max = b) - punif(205, min = a, max = b)
>
> # Print the results
> cat("P(x > 230):", prob_x_gt_230, "\n")
P(x > 230): 0.25
> cat("P(205 ≤ x ≤ 220):", prob_range_205_to_220, "\n")
P(205 ≤ x ≤ 220): 0.375
>
```

Solution:

Uniform distribution between 200 and 240



i. P(x>230)

$$P(x) = \frac{d-c}{b-a}$$

Where a=200, b=240, d=240 , c=230

$$= \frac{240-230}{240-200} = 0.25$$

ii. P(205≤x≤220)

$$P(x) = \frac{220-205}{240-200} = \frac{15}{40} = 0.37$$

7. Following are the scores of max vertical jumps before and after the training program. Test whether the training program is helpful to the students (Use Paired t-test).

Player	Max Vertical Jump Before Training Program	Max Vertical Jump After Training Program
Player 1	22	24
Player 2	20	22
Player 3	19	19
Player 4	24	22
Player 5	25	28
Player 6	25	26
Player 7	28	28
Player 8	22	24
Player 9	30	30
Player 10	27	29
Player 11	24	25
Player 12	18	20
Player 13	16	17
Player 14	19	18
Player 15	19	18
Player 16	28	28
Player 17	24	26
Player 18	25	27
Player 19	25	27
Player 20	23	24

Date: 24-04-2024

****

R-Code:

```
# Example data for "before" and "after" scores (replace with your actual data)
before_scores <-  c(22,20,19,24,25,25,28,22,30,27,24,18,16,19,19,28,24,25,25,23)
after_scores    <-  c(24,22,19,22,28,26,28,24,30,29,25,20,17,18,18,28,26,27,27,24)

# Perform the paired t-test
result <- t.test(before_scores, after_scores, paired = TRUE)

# Print the results
cat("Paired t-test results:\n")
cat("Mean difference:", result$estimate, "\n")
cat("T-statistic:", result$statistic, "\n")
cat("P-value:", result$p.value, "\n")

# Check the significance at a specific alpha level (e.g., alpha = 0.05)
alpha <- 0.05
if (result$p.value < alpha) {
      cat("The training program is statistically helpful to the students.\n")
} else {
     cat("There is no statistically significant evidence that the training program is helpful.\n")
}
```

OUTPUT:

```
> before_scores <- c(22,20,19,24,25,25,28,22,30,27,24,18,16,19,19,28,24,25,25,23)
> after_scores <-  c(24,22,19,22,28,26,28,24,30,29,25,20,17,18,18,28,26,27,27,24)
>
> # Perform the paired t-test
> result <- t.test(before_scores, after_scores, paired = TRUE)
>
> # Print the results
> cat("Paired t-test results:\n")
Paired t-test results:
> cat("Mean difference:", result$estimate, "\n")
Mean difference: -0.95
> cat("T-statistic:", result$statistic, "\n")
T-statistic: -3.226173
> cat("P-value:", result$p.value, "\n")
P-value: 0.004445371
>
> # Check the significance at a specific alpha level (e.g., alpha = 0.05)
> alpha <- 0.05
> if (result$p.value < alpha) {
+   cat("The training program is statistically helpful to the students.\n")
+ } else {
+   cat("There is no statistically significant evidence that the training program is helpful.\n")
+ }
The training program is statistically helpful to the students.
```

Solution:

Player	Max vertical jump before training (A)	Max vertical jump after training (B)	d=A-B	d-d' (d'=∑d /n d'= -0.95)	$(d-d')^2$
1	22	24	-2	-1.05	1.102
2	20	22	-2	-1.05	1.102
3	19	19	0	0.95	0.902
4	24	22	2	2.95	8.702
5	25	28	-3	-2.05	4.202
6	25	26	-1	-0.05	0.002
7	28	28	0	0.95	0.902
8	22	24	-2	-1.05	1.102
9	30	30	0	0.95	0.902
10	27	29	-2	-1.05	1.102
11	24	25	-1	-0.05	0.002
12	18	20	-2	-1.05	1.102
13	16	17	-1	1.95	0.002
14	19	18	1	1.95	3.802
15	19	18	1	1.95	3.802
16	28	28	0	0.95	0.902
17	24	26	-2	-1.05	1.102
18	25	27	-2	-1.05	1.102
19	25	27	-2	-1.05	1.102
20	23	24	-1	-0.05	0.002
			∑d=-19		$\sum(d-d')^2$ =32.95

$d' = \sum d / n = -19/20 = -0.95$

Sample variance $=\sum(d-d')^2/(n-1)$

$=32.95/20-1$

$=1.734211$

Sample standard deviation $S_d = \sqrt{\sum(d-d')^2/(n-1)}$

$= \sqrt{1.734211}$

$= 1.316894$

Therefore $t = d'/(s_d/\sqrt{n})$

$= -0.95/((1.316894)/\sqrt{20})$

$= -0.95/0.29446$

$= \underline{-3.226}$

Calculate t-statistic $t = \frac{\bar{d}}{s/\sqrt{n}}$

Calculate the critical value:

Degree of freedom= n-1 = 20-1 = 19

$\alpha=0.05$

$\underline{t_{(0.05,19)} = -1.729}$

-3.226 < -1.729 ➔ So reject the null hypothesis.

The training program is statistically helpful to the students.

****

8. A company has three manufacturing plants, and company officials want to determine whether there is difference in the average age of workers at the three locations. The following data are the ages of five randomly selected workers at each plant. Perform a one-way ANOVA to determine whether there is a significant difference in the mean ages of the workers at three plants. Use α=0.01. Write R script for the above problem.

Plant(Employee Ages)

1	2	3
29	32	25
27	33	24
30	31	24
27	34	25
28	30	25

Date: 24-04-2024

****

R Code:

```
# Create a data frame with the ages of workers at each plant
data <- data.frame(Plant = factor(rep(1:3, each = 5)),
Employee_Ages = c(29, 27, 30, 27, 28, 32, 33, 31, 34, 30, 25, 24, 24, 25, 25))

# Perform a one-way ANOVA
anova_result <- aov(Employee_Ages ~ Plant, data = data)

# Summarize the ANOVA results
sum_res<- summary(anova_result)

# Print the ANOVA table and check for significance
cat("One-way ANOVA results:\n")
sum_res
p_value<-sum_res[[1]][[5]]


# Check the significance at alpha = 0.01
alpha <- 0.01
if (p_value[1] < alpha) {
    cat("There is a significant difference in the mean ages of workers at the three plants.\n")
} else {
   cat("There is no significant difference in the mean ages of workers at the three plants.\n")
}
```

OUTPUT:

```
# Create a data frame with the ages of workers at each plant
> data <- data.frame(Plant = factor(rep(1:3, each = 5)),
+   Employee_Ages = c(29, 27, 30, 27, 28, 32, 33, 31, 34, 30, 25, 24, 24, 25, 25))
>
> # Perform a one-way ANOVA
> anova_result <- aov(Employee_Ages ~ Plant, data = data)
>
> # Summarize the ANOVA results
> sum_res<- summary(anova_result)
>
> # Print the ANOVA table and check for significance
> cat("One-way ANOVA results:\n")
One-way ANOVA results:
> sum_res
            Df Sum Sq Mean Sq F value   Pr(>F)
Plant        2  136.9    68.47   45.64 2.46e-06 ***
Residuals   12   18.0     1.50
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
> p_value<-sum_res[[1]][[5]]
>
> # Check the significance at alpha = 0.01
> alpha <- 0.01
> if (p_value[1] < alpha) {
+   cat("There is a significant difference in the mean ages of workers at the three plants.\n")
+ } else {
+   cat("There is no significant difference in the mean ages of workers at the three plants.\n")
+ }
There is a significant difference in the mean ages of workers at the three plants.
```

Solution:

P1	**P2**	**P3**
29	32	25
27	33	24
30	31	24
27	34	25
28	30	25
X1’=∑141/5=28.2	X2’=∑160/5=32.0	X3’=∑123=24.6

$df_c = 3-1 = 2$

$df_e = n - c = 15 - 3 = 12$ # n = 15 c = 3

x’ = (X1’+X2’+X3’)/3 = (28.2+32.0+24.6)/3 = 28.26

Sum of squares column SSC=$\sum_{j=1}^{c} nj(x'_j - x')^2$

$= 5(28.2 - 28.26)^2 + 5(32 - 28.6)^2 + 5(24.6 - 28.26)^2$

$= 5(-0.06)^2 + 5(3.74)^2 + 5(-3.66)^2$

= 5(0.00036) + 5(13.98) + 5(13.39)

= 0.018 + 69.9 + 66.95

= 136.868

Sum of squares error SSE= $\sum^{n}{}_{i=1} \sum_{j=1}^{c} (x_{ij} - x_j')^2$

SSE = $(29-28.2)^2 + (27-28.2)^2 + (30-28.2)^2 + (27-28.2)^2 + (28-28.2)^2 +$
$(32-32.0)^2 + (33-32)^2 + (31-32)^2 + (34-32)^2 + (30-32)^2 +$
$(25-24.6)^2 + (24-24.6)^2 + (24-24.6)^2 + (25-24.6)^2 + (25-24.6)^2$

= 0.64+1.44+3.24+1.44+0.04+0+1+4+4+0.16+0.36+0.36+0.16+0.16

<u>= 17.64</u>

MSC = SSC/dfc = 136.86/2 = 68.43 # n-1 = 3 col-1 = 2

MSE = SSE/dfe = 17.64/12 = 1.47 # n-15 = 15 – 3col = 12

F value = MSC/MSE = 68.43/1.47 <u>= 46.55</u>

$F_{(dfc,dfe)} = F_{(2,12)} = 6.93$ #from chart

46.55 > 6.93,

Therefore, there is a significant difference in the mean ages of workers at the three plants.